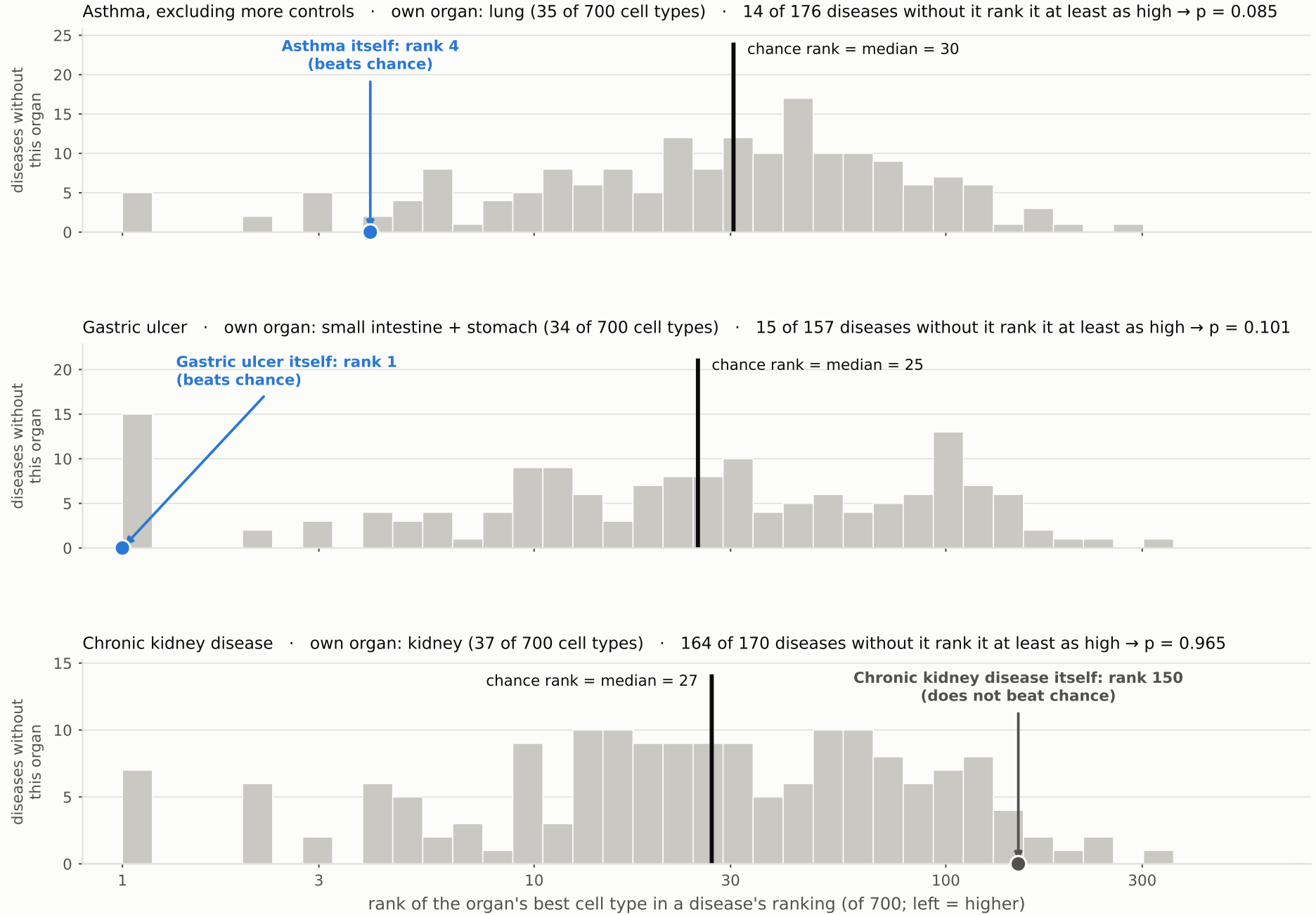
How the chance rank (black tick) and the p-value in Figures 3 and 5 are computed

Gray bars: take every disease that does NOT have this organ among its own organs, and find where the organ's best cell type lands in that disease's ranking.
Black line: the median of those positions = the organ's chance rank. The disease beats chance if its own rank (dot) is left of the black line.
p = (1 + gray diseases at or left of the dot) / (1 + all gray diseases): how unusual the disease is against the whole spread, not just the median.